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Systematic investigation of fabrication conditions of nanocarrier based on carboxymethyl chitosan for sustained release of insulin

Xiaoyu Bai[a,1], Ming Kong[a,1], Guixue Xia[a], Shichao Bi[a], Zhongzheng Zhou[a], Chao Feng[a], Xiaojie Cheng[a,**], Xiguang Chen[a,b,*]

[a] College of Marine Life Science, Ocean University of China, Qingdao 266003, PR China
[b] Qingdao National Laboratory for Marine Science and Technology, Qingdao 266000, PR China

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ABSTRACT

pH-responsive nanoparticles (NPs) comprised of degradable carboxymethyl chitosan (CMCS) crosslinked with $CaCl_2$ were simply prepared *via* ionic gelation. Fabrication conditions including insulin dosage, CMCS concentration, and crosslinking density were systematically investigated for insulin loading and release *in vitro*. The encapsulation efficiencies (EE), loading capacity (LC) and average size of the NPs decreased with the increasing insulin concentrations (<0.192 mg/mL), while they notably increased as the insulin dosage was above 0.192 mg/mL. When the concentration of CMCS increased from 0.5 to 2.0 mg/mL, the EE of the NPs reduced while the size of the NPs increased. We further demonstrated that crosslinking density offered a simple method for tuning the properties of the NPs towards various insulin concentrations. The mass ratio 10:5 of CMCS to $CaCl_2$ exhibited the optimal performance at higher insulin concentration, whereas a higher crosslinking density of 10:7 (m:m) gave the optimal performance at low insulin concentration. The cumulative release of insulin from insulin loaded NPs decreased with the elevating crosslinking density. These findings not only provided a better understanding of the synthesis of CMCS NPs but also contributed to the practical applications of insulin loading and release.

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1. Introduction

Insulin is the most commonly used and effective drug for controlling plasma glucose levels. Insulin therapy is generally delayed despite the dire consequences, and partly due to the inconvenience and complications associated with insulin administration by injection [1]. A recent study showed that over half of users reported intentionally skipping insulin injections at some point, and 20% reported regularly skipping injections, leading to a dangerous lapse in treatment [2]. Hence, in the last decade, the research focus has shifted from the development of insulin alternatives to the pursuing of alternative delivery methods [3–6]. The use of biocompatible and biodegradable nanoparticles has been described as a promising strategy towards oral administration of proteins and peptides[7].

Chitosan is a linear polysaccharide composed of randomly distributed β-(1–4)-linked D-glucosamine and N-acetyl-D-glucosamine and produced by the alkaline deacetylation of chitin [8–10]. Chitosan and its derivatives possess excellent biocompatibility, biodegradability, non-toxic and hydrophility, which render them promising candidates of oral insulin delivery [11,12]. Carboxymethyl chitosan (CMCS) has carboxymethyl groups on hydroxyl sites of the glucosamine units of the chitosan structure, which has desirable water-solubility besides the advantageous biological properties of chitosan. CMCS based nanoparticles (NPs) have been extensively studied for delivery of anti-cancer agents, therapeutic proteins, genes and antigens [13–16]. It has been reported that carboxymethyl chitin NPs could be easily formed with Ca^{2+} through simple ionic gelification [17], and the carboxyl groups of CMCS make its own similar capacity of binding Ca^{2+} [18]. Moreover, many studies have emphasized the significance of concentration of chitosan in the NPs loading capacity [8,19,20].

We have previously demonstrated that CMCS NPs with negative surface charges could improve insulin delivery capability, which exhibited a higher mucoadhesion in small intestine *in ex vivo* studies, and could regulate the blood glucose level more effectively, compared to the chitosan NPs with positive surface charges

* Corresponding author at: College of Marine Life Science, Ocean University of China, Qingdao 266003, PR China.
** Corresponding author.
E-mail addresses: xjcheng@ouc.edu.cn (X. Cheng), xgchen@ouc.edu.cn (X. Chen).
[1] These two authors are equal as the first author in this work.

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Table 1
Formulation of synthetic CMCS NPs sat different conditions.

Sample	CMCS (mg/mL)	CMCS:$CaCl_2$ (m:m)
1	2	10:5
2	1	10:5
3	0.75	10:5
4	0.5	10:5
5	0.5	10:1
6	0.5	10:3
7	0.5	10:7

[21,22]. In this work, we extend the previous work to systematically study the effects of fabrication conditions of CMCS NPs, including insulin dosage, CMCS concentration, and crosslinking density, on the insulin loading capacity. The encapsulation efficiencies (EE), loading capacity (LC), size, surface charge and release profile of these NPs were evaluated. Furthermore, the effect of preparation conditions on insulin release under different pH conditions was also comprehensively studied.

2. Materials and methods

2.1. Materials

CMCS (weight-average molecular weight (MW): 190 kDa, degree of deacetylation (DD): 85.31%, degree of substitution (DS) of carboxymethyl: 0.95) was obtained from our laboratory using the method described by Chen [23,24] previously. Insulin was purchased from Sigma (St. Louis, USA). All other reagents and solvents were of analytical grade.

2.2. Preparation of CMCS nanoparticles (NPs)

CMCS NPs were synthesized by a simple ionic gelation method with calcium chloride ($CaCl_2$) solution [25]. In brief, CMCS was dissolved in triple distilled water at predetermined concentration, and insulin was dissolved in hydrochloric acid solution (pH 2). 1 mL of insulin solution was added into 8 mL of CMCS solution dropwise under constant stirring. Subsequently, 1 mL $CaCl_2$ solution was added into the mixture dropwise under constant stirring for 1 h until NPs formation. The same procedure was applied to the preparation of NPs at different concentrations of CMCS and mass ratio of CMCS to $CaCl_2$ was listed in Table 1.

2.3. Size and surface charge characterizations of the NPs

The size and surface charge of blank NPs and insulin loaded NPs were characterized and measured by dynamic light scattering (DLS) with a Malvern Zetasize Nano ZS 90 (Malvern Instruments Ltd., Malvern, UK) at a detector angle of 90, 670 nm, and 25 °C. The samples were prepared with deionized water at appropriate concentrations. All measurements were performed in triplicate.

2.4. Transmission electron microscopy (TEM)

The shapes of the insulin free and insulin loaded NPs were observed by transmission electron microscope (TEM). The CMCS NPs suspensions were stained by phosphotungstic acid, placed onto copper grid and dried at room temperature, then observed by TEM (100 CX II, Japan).

2.5. Encapsulation efficiencies (EE) and loading capacity (LC) of insulin

The EE refers to the amount of insulin associated with formed NPs, expressed as the percentage of the total amount of insulin added in the process, while LC is defined as the percentage by the weight of the associated insulin in the resultant NPs. To determine these parameters, NPs were separated by centrifugation (15,000 rmp/min, 30 min and 4 °C) from the aqueous preparation medium containing the free insulin. And the free insulin content was estimated by BCA method at 562 nm in a microplate reader (MCC340, Multiskan, Belgium) [26]. The EE and LC were calculated using the following formulas [27]:

$$\text{EE}(\%) = ((\text{total amount of insulin added-free insulin}) / (\text{total amount of insulin added})) \times 100\%$$

$$\text{LE}(\%) = ((\text{total amount of insulin added-free insulin}) / (\text{weight of nanoparticles})) \times 100\%$$

2.6. Cytotoxicity

The cytotoxicity of the NPs was evaluated using MTT assay conducted with colon adenocarcinoma (Caco 2) cell line and L929 cell line, which were obtained from the American Type Culture Collection (Manassas, USA, between 25 and 35 passages). The cells were cultured in Dulbecco's modified Eagle's medium (DMEM; high glucose) and supplemented with 10% (v/v) fetal bovine serum and penicillin/streptomycin at 37 °C in 5% CO_2. Briefly, cells in the logarithmic growth phase were added to 96-well culture plates of 5×10^4 cells/mL at 100 μL per well and incubated overnight. The culture medium was then replaced by the serum-free medium with various concentrations (500 μg/mL, 250 μg/mL, 125 μg/mL, 62.5 μg/mL and 31.25 μg/mL) of insulin loaded CMCS NPs, and MTT assays were performed at 24 h, 48 h and 72 h after the addition of the NPs dilutions. Cells without the NPs treatment were as control. Cell viability (%) was calculated as follows[28]:

$$\text{cell viability}(\%) = \left(\text{OD}_{\text{test}}/\text{OD}_{\text{control}}\right) \times 100\%$$

Where OD_{test} and $\text{OD}_{\text{control}}$ are the absorbance of the tested sample and the control sample.

2.7. In vitro drug release

Insulin released from NPs was analyzed by incubating NPs in PBS buffer solutions at pH 1.2 and pH 6.8 media. NPs suspension (5 mL) was placed into a cellulose membrane dialysis tube (MW cutoff 8000–14,000). The dialysis tube was placed in 45 mL of PBS at pH 1.2 and pH 6.8 at 37 °C while shaking (100 rpm), respectively. At specific time intervals (between 0 h and 4 h), 1 mL medium was removed and replaced by 1 mL fresh release medium. The amount of insulin released in the medium was determined by BCA method at 562 nm in a microplate reader. And a calibration curve was generated using blank NPs to correct for the intrinsic absorption of the NPs.

2.8. Statistical analysis

All the results were expressed as mean ± SD, n = 3. The significance level was determined by one-way ANOVA following Tukey'spost hoc test. $p < 0.05$ was considered as significant.

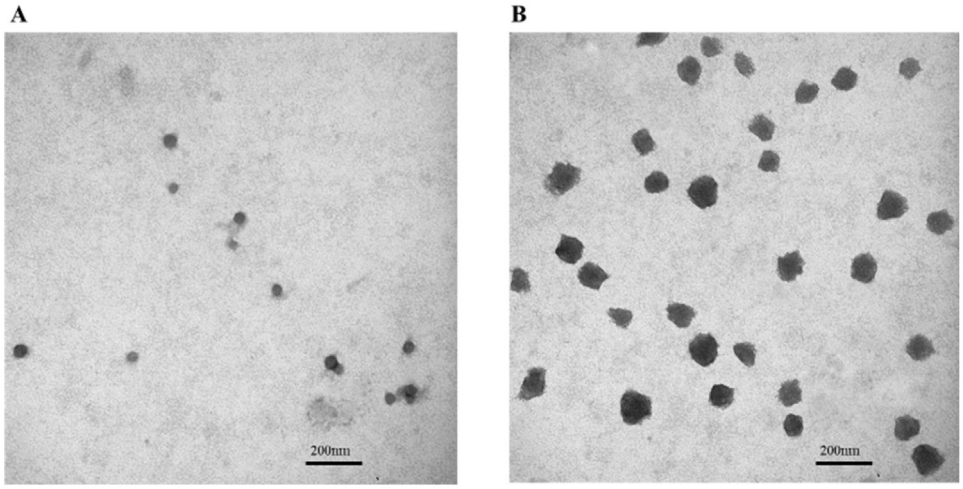

Fig. 1. TEM of CMCS NPs (A) and insulin loaded CMCS NPs (B). NPs formulation: CMCS concentration:0.5 mg/mL, CMCS:$CaCl_2$ (m:m):2:1, insulin concentration:0.528 mg/mL, pH 6.5.

Table 2
Effect of insulin concentrations on particles average size, polydispersity and surface charge of the NPs. NPs formulation: CMCS concentration:0.5 mg/mL, CMCS: $CaCl_2$ (m:m):2:1, pH 6.5.

Insulin concentration(mg/mL)	0	0.024	0.048	0.096	0.192	0.240	0.336	0.432	0.528
Size(nm)	74.68	90.30	85.71	83.28	83.39	85.25	88.20	118.30	123.71
PDI	0.028	0.111	0.014	0.047	0.002	0.030	0.070	0.128	0.152
Zeta(mV)	−12.4	−11.4	−13.2	−14.1	−7.1	−14.4	−14.4	−16.2	−16.2

3. Results and discussion

3.1. Preparation and the characterization of CMCS NPs

CMCS NPs with different CMCS concentration and crosslinking density were successfully prepared by the ionic gelation between positively charged Ca^{2+} and negatively charged CMCS (Table 1). As shown in Fig. 1, the obtained nanoparticle suspension displayed a spherical morphology with smooth surfaces and uniform size distributions. The insulin loaded NPs (Fig. 1B) presented a larger diameter of 123.71 nm than the corresponding insulin free NPs of 74.68 nm (Fig. 1A) due to the loading of insulin. The size of the nanoparticles shown in TEM images was much smaller than that detected by DLS (Table 2) owing to the drying process in TEM sample preparation.

3.2. Effect of insulin concentration on the loading capacity of nanoparticles

Since insulin are protein molecules with complex 3-D structure, and are able to fold and unfold, their interactions with the long chain of CMCS and the consequential encapsulation can be complicated, which mostly depends on 3-D conformation, electrostatic, and solution conditions [29]. Therefore, insulin concentration would affect the EE, LC, size, and surface charge of the NPs. Step increase of insulin concentration from 0.024 to 0.192 mg/mL during the encapsulation process resulted in the decreasing EE from 45.6% to 4.4% and LC from 2.6% to 2.0%, while further increase of insulin concentration led to an almost linearly increasing EE from 4.4% to 60.41% and LC from 2.0% to 44.36% (Fig. 2).

Protein encapsulation *via* electrostatic interactions typically exhibits a saturation kinetics, which reaches to peak value at certain concentrations under constant temperature [19]. In the preparation process of insulin loaded CMCS NPs, the acid solution (pH 2) conferred insulin with positive charges, since the pI of insulin is 5.3, which were rapidly neutralized by CMCS solution. In the process of NPs formation, the electronic interaction occurred instantly between the carboxyl groups of CMCS and certain amount of insulin ahead of charge conversion. While the left insulin molecules with converted charges could still associate with CMCS *via* Ca^{2+} crosslink but within quite limited extent due to the electrostatic repulsion. At the lower insulin concentrations from 0.024 to 0.192 mg/mL, the electrostatic repulsion highly hampered the association of insulin molecules to CMCS, leading to saturation of the insulin encapsulation. Consequently, continuing increase of insulin concentration would nearly have no effect on the loading (LC) of NPs and reduce the encapsulation (EE) of the NPs. Inversely, EE of the NPs increased at the higher insulin concentrations from 0.24 to 0.528 mg/mL. The phenomena might be attributed to two likelihoods. First, as the insulin concentration increased, insulin molecules combined with Ca^{2+} to form aggregates that might precipitate during cen-

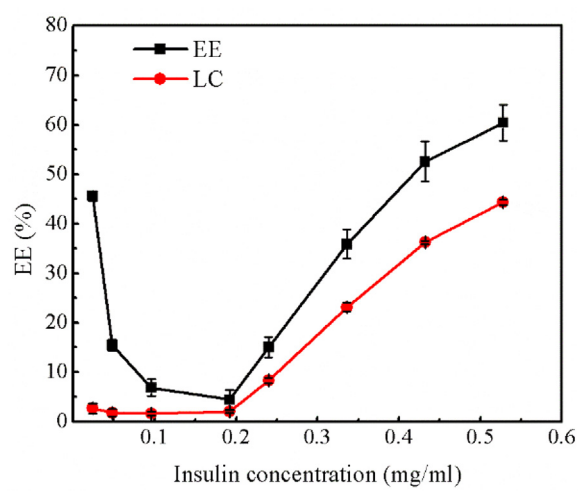

Fig. 2. Effect of insulin concentrations on EE and LC in the NPs. NPs formulation:CMCS concentration:0.5 mg/mL, CMCS:$CaCl_2$ (m:m):2:1, pH 6.5.

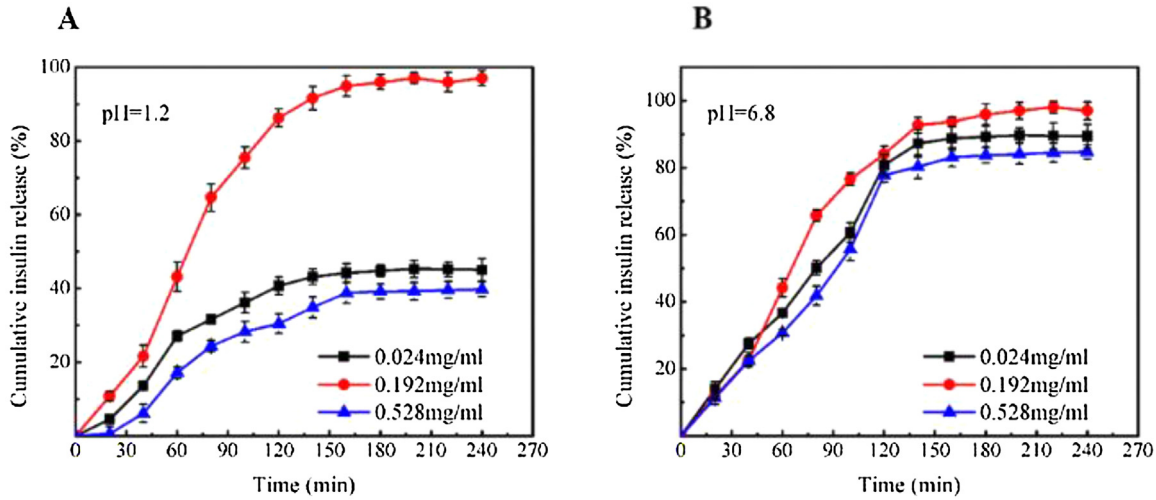

Fig. 3. Effect of insulin concentration on subsequent insulin release profile. NPs formulation:CMCS concentration:0.5 mg/mL, CMCS:$CaCl_2$ (m:m):2:1, pH 6.5.

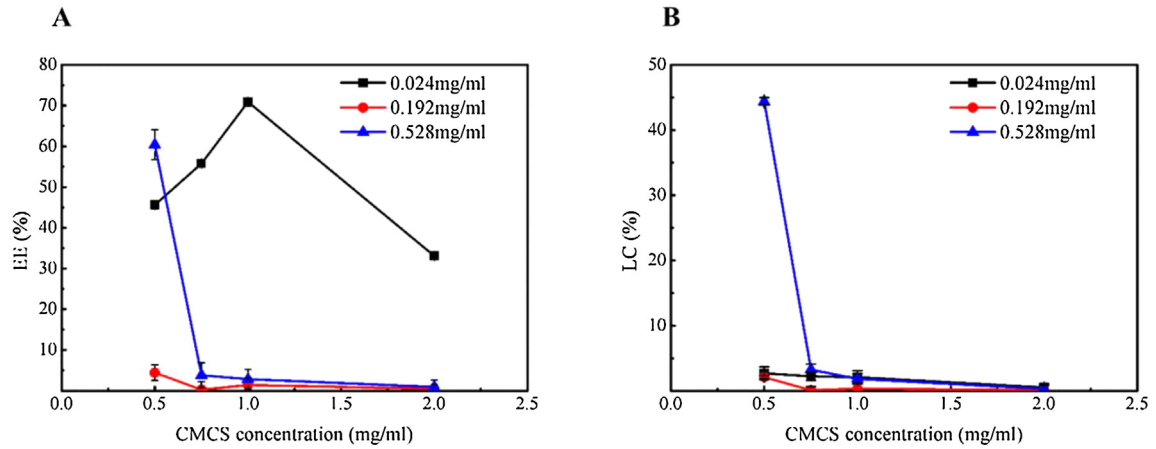

Fig. 4. Effect of CMCS concentration on EE and LC of NPs. NPs formulation: CMCS:$CaCl_2$ (m:m):2:1, pH 6.5.

trifugation leading to the increase of EE. For another, the insulin aggregates might be deposited or attached on the surface of CMCS NPs *via* Ca^{2+}, thus resulting in the increase of insulin incorporation, similar to the multilayer protein adsorption on surfaces at higher concentration range above monolayer saturation concentration [19,30–33]. In order to distinguish the likelihoods, same concentrations of insulin (0.528 mg/mL) and $CaCl_2$ were blended to form aggregates but no precipitation was obtained after centrifugation. Furthermore, there is no change of the detected amount of insulin in solution with centrifugation or not. The results indicated that the aggregates of insulin and Ca^{2+} was not large enough to be precipitated by themselves during centrifugation. However, the insulin aggregates formed at high concentration probably could deposit or be attached on the surface of CMCS NPs *via* Ca^{2+}.

The encapsulation of insulin showed a remarkable increase in particle size from 74.68 nm to 123.71 nm and slight variation in surface charge (Table 2). As insulin concentration ranging from 0.024 mg/mL to 0.192 mg/mL, the average size of the NPs reduced with the decline of LC, namely, from 90.30 nm at 3.62% to 83.39 nm at 1.62%. As insulin concentration was above 0.192 mg/mL, the increase of LC from 2.08% to 44.36% resulted in larger average size of the NPs from 85.25 nm to 123.71 nm. Moreover, the decrease of zeta potential evidenced the attachment of insulin onto the surface of nanoparticles. Notably, at insulin concentration of 0.192 mg/mL, the NPs had the lowest EE and LC and exhibited the smallest amount of surface charges due to the neutralization of charges. This result verified the mechanisms we speculated at this concentration that the strong electrostatic repulsion between CMCS and insulin induced the lowest EE and the least quantity of surface charges.

The release profile of the NPs with different insulin concentrations were evaluated under different pH condition. After a 240-min dialysis against PBS, the cumulative release of insulin maintained 39–97% at pH 1.2 and 77–98% at pH 6.8, respectively, shown in Fig. 3. This can be explained by the fact that some insulin molecules were cross-linked with the CMCS network by Ca^{2+} in the NPs. Meanwhile, the release of insulin was partially inhibited at pH 1.2 for all tested NPs compared with pH 6.8. The higher pH of the PBS, the larger amount of carboxylic acid salt ($-COO^-$) led to increased electrostatic repulsion of inter- and intra-molecules and larger pore size, which facilitated release of insulin. PBS of pH 1.2 and 6.8 simulated the pH and ion condition in gastro and small intestine, respectively. As expected, the release of insulin from the NPs could be suppressed in gastro (pH 1.2) and promoted as long as the NPs entered in small intestine (pH 6.8). For insulin concentration of 0.192 mg/mL, the released amount of insulin was significantly higher than the others both at pH 1.2 and pH 6.8, probably ascribing from the association between insulin and CMCS was the weakest. Additionally, the cumulative release of insulin decreased with the increase of the insulin concentrations, higher concentrations of insulin were more likely to accelerate the hydrophobic interactions to self-aggregate

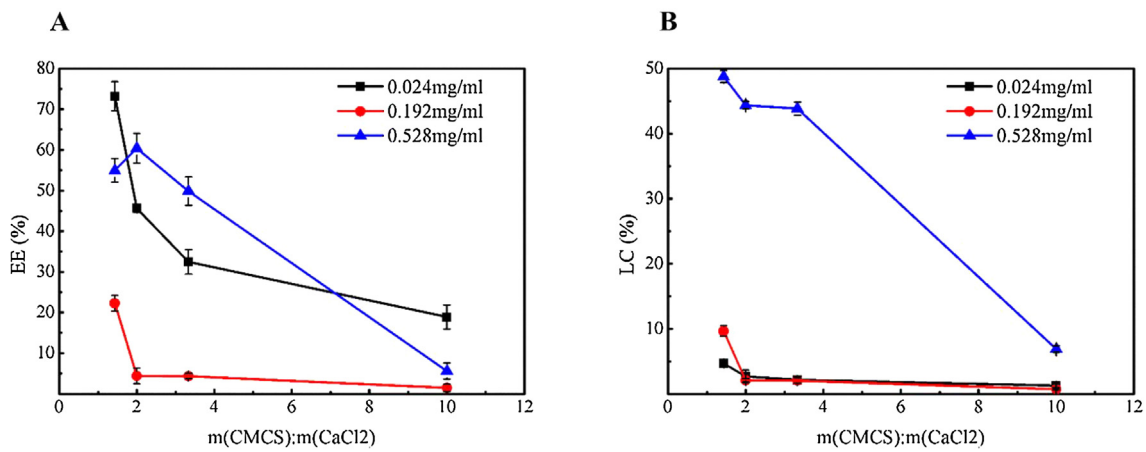

Fig. 5. Effect of CMCS to $CaCl_2$ mass ratios on EE (A) and LC (B) of NPs. NPs formulation:CMCS concentration:0.5 mg/mL, pH 6.5.

into a critical nucleus and eventually into fibrils for insulin, thus decreasing the cumulative release of insulin from the NPs [31].

3.3. Effect of CMCS concentration on the loading capacity of nanoparticles

The effect of CMCS concentrations (0.5 mg/mL to 2 mg/mL) on the insulin EE and LC of the NPs during the process of preparation was studied. At different insulin concentrations (0.024 mg/mL, 0.192 mg/mL and 0.528 mg/mL), the increase of CMCS concentrations rendered the EE raise from 0.17% to 70.89% (Fig. 4A), and the LC from 0.05% to 44.36% (Fig. 4B).

Obviously, the concentration of CMCS has a significant influence on its loading capacity of insulin. For insulin of 0.024 mg/mL, the EE and LC of the NPs notably increased with the concentration of CMCS from 0.5 to 1.0 mg/mL, whereas further increasing CMCS concentration led to an opposite effect on EE from 70.89% to 33.09% (Fig. 4A). At low insulin concentration, the association of insulin to CMCS was hampered by the electrostatic repulsion. Within concentration of CMCS from 0.5 to 1 mg/mL, more CMCS could associate more insulin molecules ahead of charge conversion. What if the concentration of CMCS increased further, the rising solution viscosity would hinder encapsulation of insulin by deterring insulin molecule movement around CMCS molecular chain [34]. For insulin of 0.192 mg/mL, the EE was the lowest in all studied CMCS concentrations that is coincide with the above results.

For insulin of 0.528 mg/mL, the insulin EE of the NPs decreased dramatically from 60.41% to 1% with increasing CMCS concentration. When the concentration of CMCS increased, CMCS predominated in the solution, and Ca^{2+} preferentially bonded to the CMCS macromolecules chains. Therefore, the Ca^{2+} required for insulin aggregates formation and adsorption of the insulin aggregates onto the surface of CMCS NPs were insufficient, leading to the reduction of NPs EE at high concentration of CMCS. It is reasonable to believe that the concentration of CMCS will play a major role in association of insulin to CMCS NPs.

3.4. Effect of mass ratio (CMCS to $CaCl_2$) on the loading capacity of nanoparticles

In the synthesis of NPs, the crosslink density of the NPs is directly proportional to the mass ratio of CMCS to cross-linker agent $CaCl_2$ [33]. The effects of CMCS to $CaCl_2$ mass ratio on insulin encapsulation, particles average size, polydispersity, surface charge of NPs and subsequent release were studied at the mass ratios of 10:1, 10:3, 10:5, and 10:7, with the fixed CMCS concentration of 0.5 mg/mL.

When the mass ratio of CMCS to $CaCl_2$ was 10:1, the NPs have the lowest the EE and LC (Fig. 5A). It implied that the majority of CMCS molecules might exist as free polymers or oligomers, and the reaction solutions were clear without visible opalescence, indicating that the $CaCl_2$ concentration was inadequate to cross-link CMCS leading to the lowest EE and LC.

With the increase of mass ratios of CMCS to $CaCl_2$, the insulin EE of the NPs decreased at insulin concentration of 0.024 and 0.192 mg/mL (Fig. 5A). The insulin LC of the NPs decreased with the increase of CMCS to $CaCl_2$ mass ratios (Fig. 5B). At the higher mass ratio of the CMCS to $CaCl_2$, namely reducing the $CaCl_2$ concentration would reduce the cross-linking density of insulin loaded CMCS NPs, giving rise to a more loose structure of CMCS NPs to entrap less insulin and resulting in the low EE.

At the insulin concentration of 0.528 mg/mL, Ca^{2+} not only crossly linked CMCS chains into nanoparticles but also bind insulin to form aggregates adsorbed on CMCS nanoparticles. Therefore, with the increase of Ca^{2+} concentrations, the NPs have a higher loading level. However, different from the other concentrations of insulin, when the ratio of CMCS to $CaCl_2$ was 10:7, the EE of the NPs decreased. It is assumed that Ca^{2+} would occupy the surface binding sites of the NPs, hindering the combination of insulin aggregates to CMCS NPs. It was also observed that the NPs would aggregate if the mass ratio of CMCS to $CaCl_2$ decreased less than 10:7. The reason behind this behavior may be that the low surface charge density of the particles was no longer able to maintain the stability of these large particles during stirring, which resulting in the precipitation of particles. It concluded that the concentration of $CaCl_2$ was the crucial parameter for the association of insulin to CMCS NPs.

To test the effect of crosslinking density on the particle size, polydispersity and surface charge of the NPs, a series of NPs were also prepared with different mass ratios of CMCS to $CaCl_2$. With the decrease of CMCS to $CaCl_2$ mass ratios from 10:1–10:7, the particle sizes were initially decreased dramatically and then increased at the studied insulin concentration, where the PDI values were significantly decreased (Table 3), which is consistent with the variation of LC of the NPs.

With the decreasing mass ratios, the enhancement of the crosslinking density of the NPs led to a more compact structure and a narrower size distribution. Further decreasing the mass ratio could increase the insulin LC of the NPs and the particle sizes. Our inference could be proved by the results that CMCS molecules were almost entirely cross-linked, and the excessed $CaCl_2$ would combine with more insulin form aggregates that adsorbed onto the

Table 3
Effect of CMCS to $CaCl_2$ mass ratios on particles average size, polydispersity and surface charge of NPs. NPs formulation:CMCS concentration:0.5 mg/mL, pH 6.5.

m(CMCS):m($CaCl_2$)	insulin concentration 0.024 mg/mL				insulin concentration 0.192 mg/mL				insulin concentration 0.528 mg/mL			
	10:1	10:3	10:5	10:7	10:1	10:3	10:5	10:7	10:1	10:3	10:5	10:7
Size(nm)	170.41	153.90	90.30	91.61	169.50	152.20	83.39	90.60	164.90	76.26	123.71	121.10
PDI	0.349	0.195	0.111	0.097	0.233	0.171	0.002	0.091	0.363	0.166	0.152	0.091
Zeta(mV)	−17.2	−13.1	−11.4	−14.8	−18.1	−9.6	−7.2	−13.6	−17.9	−16.3	−16.2	−16.0

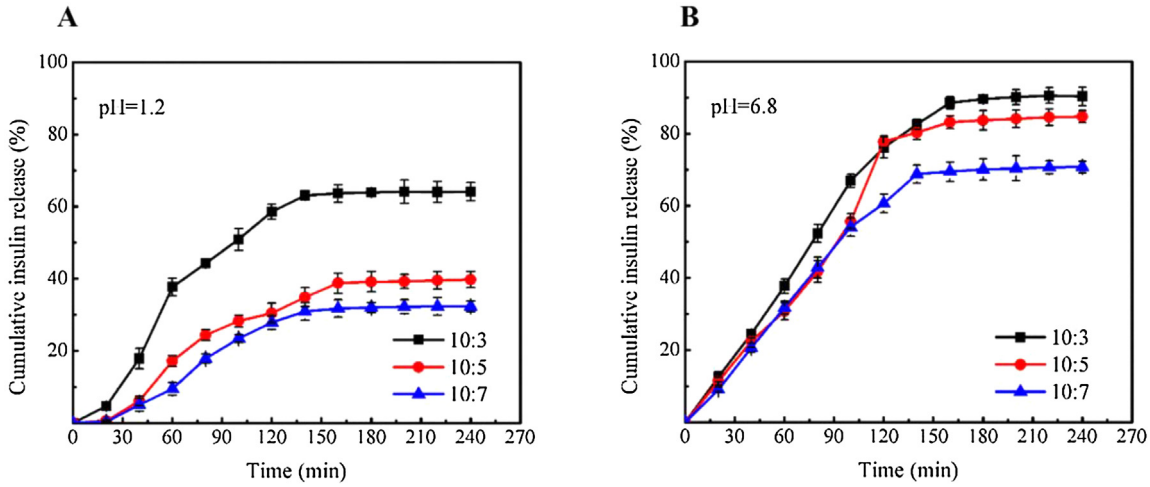

Fig. 6. Effect of CMCS to $CaCl_2$ mass ratios on subsequent insulin release profile. NPs formulation: CMCS concentration:0.5 mg/mL, insulin concentration:0.528 mg/mL, pH 6.5.

nanoparticle and induced larger particle size. Within the mass ratio ranges from 10:3–10:7, there was no regular variation to surface charge, and it was attributed to many factors including concentration of insulin, stability of DLS instrument and others. As expected, modulation of the crosslinking density and the concentration of insulin would directly affect the physicochemical differences of NPs, which have significant biological implications such as in the cellular uptake and biological processes [14,35].

To test the effect of crosslinking density on insulin release capability, a release experiment was conducted using insulin loaded CMCS NPs, which were prepared using CMCS to $CaCl_2$ mass ratio (m: m) of 10:3, 10:5, and 10:7. It revealed that the cumulative amount of insulin released from the NPs of CMCS to $CaCl_2$ 10:3 was the highest, which was about 64% in pH 1.2 and approximately 90% in pH 6.8 (Fig. 6A). For the NPs of CMCS to $CaCl_2$ 10: 5 and 10:7, the released amount of insulin was significantly lower than the NPs of CMCS to $CaCl_2$ 10: 3 in pH 1.2. In pH 6.8, the NPs of CMCS to $CaCl_2$ 10:3, 10:5, and 10:7 had the same release rate in the first hour. However, the NPs of CMCS to $CaCl_2$ 10:7 had the lowest cumulative release amount about 60% after 2 h. Moreover, the cumulative release of insulin from the NPs decreased with the decreasing mass ratio of CMCS to $CaCl_2$. The reason was attributed to the high $CaCl_2$ concentration which could increase the crosslinking density of insulin loaded NPs, resulting in a more compact structure of insulin loaded NPs, and decreasing the diffusion as well as cumulative release of insulin from insulin loaded NPs [36].

3.5. Cellular compatibility

This study aimed to evaluate the cellular compatibility of the CMCS NPs with L929 cells (Fig. 7A) and Caco 2 cells (Fig. 7B). The cells viability was all over 80% after treated with CMCS NPs at scheduled time intervals. It demonstrated that all tested concentrations of CMCS NPs from 500 μg/mL to 31.25 μg/mL had no obvious toxic effects on the cellular viability. The results showed that the interactions among the NPs with different concentrations had no influence on the cytotoxicity for Caco 2 cells and L929 cells. These results were similar to the report by Anitha [37].

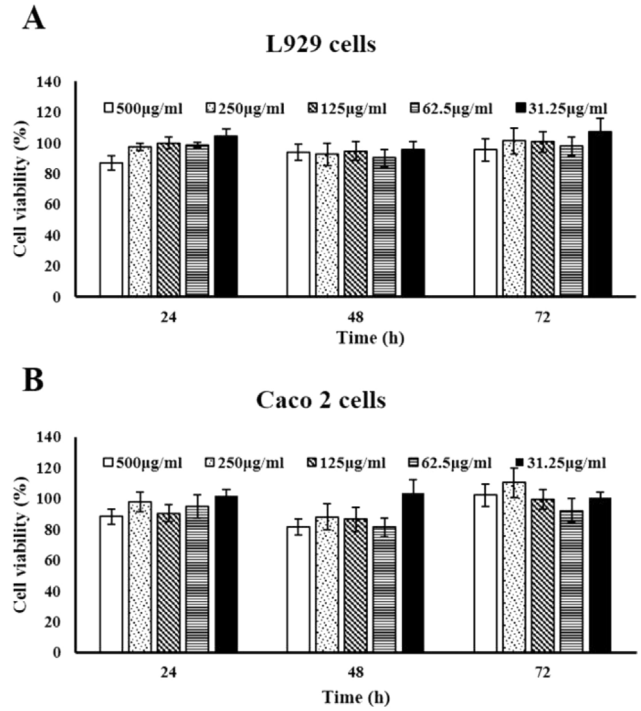

Fig. 7. *In vitro* cytotoxicity of CMCSNPs on L929 cells (A) and Caco 2 cells (B). The incubation times were 24 h, 48 h and 72 h; Data represent the mean ± SD (n = 5).

4. Conclusions

In this work, the nanoparticles (NPs) comprised of degradable CMCS cross-linked with $CaCl_2$ were successfully prepared and studied for *in vitro* protein loading and releasing. It was found that when the concentration of insulin increased, the EE, LC and diameter of the NPs decreased firstly and then increased. And the EE and LC of the NPs changed much with increasing concentration of CMCS. Besides, the decreasing mass ratio of CMCS to $CaCl_2$ led to an increased EE and size of the NPs. The cumulative release of insulin from insulin loaded NPs decreased with the elevating crosslinking density. With all indications, the optimal preparation condition is when the concentration of CMCS and insulin are 0.5 mg/mL and 0.528 mg/mL, respectively, and CMCS to $CaCl_2$ mass ratios is 10:5. This investigation will provide the rational direction for the design of nanoparticle-based insulin oral delivery vehicles and will motivate more efforts on exploring CMCS/$CaCl_2$ NPs in the applications of insulin delivery.

Authors' contributions

The manuscript was written through contributions of all authors.

Conflict of interest

The authors declare that they have no conflict of interest.

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